

Pushing the Pi5 Cluster

An empirical investigation of the bit-exact ceiling

Findings from 4 background Linux research agents and ~ 20 cluster experiments

Companion to “Distributed LLM Inference on a 4-Node Raspberry Pi 5 Cluster”

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Abstract

This companion paper documents a multi-round investigation of the Raspberry Pi 5 cluster (4×Pi 5 16 GB, Cortex-A76, Gigabit Ethernet) running **distributed-llama** on Qwen3-30B-A3B Q40. Building on the original report’s 12.71 tok/s baseline and the subsequent Stage 9 work (TIER 0 sysctls) and A2 op-fusion (OP_SILU_MUL) that reached 14.011 tok/s, we conducted Rounds 3 and 4 of experiments: four parallel background research agents examined Linux Plumbers, SOSP, OSDI, EuroSys, eBPF, scheduler, and ARM-specific papers (2024–2026); we cherry-picked the Phase B async-sync foundation; implemented EEVDF per-task slice tuning via `sched_setattr`; applied multiple sysctl bundles; and tested ~ 15 distinct configurations including threaded NAPI, IRQ pinning, per-thread CPU affinity, and `-tile-sync` K-values. **Net measured improvement Rounds 3+4: 0% within noise** (Stage 11 subtotal = 14.046 tok/s).

A subsequent rigorous-bench session in May 2026 (**Stages 12–15**) added four further cumulative bit-exact commits to the production branch, all validated by SHA-256-hashing 100 deterministic tokens against the Stage 11 reference: `64ef787` removed counter-productive software prefetch from the matmul Q80×Q40 NEON kernel (HW prefetcher of the Cortex-A76 outperforms the manual hints); `1af175c` rewrote `silu_mul_F32` as a true single-pass NEON kernel (−33% memory ops); `f8ed9d2` bumped `MAX_CHUNK_SIZE` from 4 KB to 16 KB in the all-reduce loop (−4× `send()/recv()` syscalls); `edbc4ce` persisted the runtime NIC tweaks (RX ring 4096, RFS, NAPI defer) via a new `systemd` unit.

The final cluster sustains **14.449 ± 0.038 tok/s (n=20, paired, cold-start; best individual run 14.557)**, a cumulative improvement of **+5.31% over the Stage 8 baseline** of 13.720 tok/s and **+10.81% above the publicly documented ceiling of 13.04 tok/s** for this model and hardware class. The 49% backend stall rate (measured via ARM PMU `stalled-cycles-backend`) confirms the cluster remains DRAM-bandwidth-bound. Further bit-exact gains require either (a) a dedicated Async Tensor Parallelism sprint with 60-minute soak test (~ 250 LOC on top of the in-repo Phase B foundation; estimated +5–12%), (b) quality compromises (Q3 quantisation, top- $k=4$, expert pruning — explicitly out of scope of this work), or (c) hardware upgrade.

1 Background

The original technical report [1] established a 4-node Pi5 cluster running Qwen3-30B-A3B Q40 at 12.71 tok/s sustained, with 8 source-level patches to `distributed-llama`. A follow-up session (Stage 9) applied TIER 0 sysctls — GRO disable, raised TCP/socket buffers, swappiness $\rightarrow$ 1 — and pushed the cluster to 14.011 tok/s. The A2 op-fusion (OP_SILU_MUL combining the MoE FFN’s silu+mul ops with a single barrier per expert per layer) added another 0.5% bit-exact gain to 14.081 tok/s. After these gains, the question became: *can we push further under strict bit-exact, no-kernel-rebuild, no-quality-loss constraints?*

This companion note documents the Rounds 3 and 4 investigation and its empirical conclusion: **the cluster is at its bit-exact ceiling under the operational constraints.**

2 Methodology

2.1 Constraints

- Strict bit-exact (verified per-token output for a deterministic prompt at `temperature=0`).
- No kernel rebuild (production Pi OS 6.12.75).
- No reboot of the cluster (`cmdline.txt` unchanged).
- Hardware unchanged (4× Pi 5 16 GB, Gigabit Ethernet through PoE switch).
- No model quality compromise: top-*k* unchanged (8), no quantisation change, no expert pruning, no speculative decoding.

2.2 Investigation strategy

Round 3 launched four parallel background research agents:

1. **Linux Plumbers / SOSP / OSDI / EuroSys / USENIX ATC papers 2024–2026.**
2. **eBPF / XDP / AF_XDP / io_uring / kernel-bypass networking.**
3. **Linux scheduler (CFS, EEVDF in 6.6+) tunings for compute-bound workloads.**
4. **ARM-specific Linux kernel cache, prefetcher, NUMA, and memory bandwidth.**

Each agent returned ranked actionable findings. Round 4 then implemented and benchmarked the top candidates via $n = 20$ paired runs (deterministic prompt, `temperature=0`, 250-token generation, 95% CI computed).

2.3 Benchmark protocol

```
warmup: 2 discarded runs
runs:    n=20 paired
prompt:  "Write 200 words about distributed computing"
params:  max_tokens=250, temperature=0
metric:  tokens / wall-clock seconds
```

3 Results

3.1 Benchmark trajectory through the investigation

Table 1: Measured throughput across the optimisation journey (Stages 1–15, $n=20$ paired per stage).

Stage	Mean (tok/s)	±95% CI	Cumulative
b4rtaz #255 public ceiling (Pi5 8GB)	13.04	—	baseline
Stage 1–8 baseline	13.720	0.050	+5.2%
Stage 9 — TIER 0 sysctls (GRO off, buffers, swap)	14.011	0.051	+7.45%
A2 — OP_SILU_MUL fusion (bit-exact)	14.081	0.055	+7.99%
Round 3a — additional sysctls (autogroup, autocorking, etc)	14.108	0.056	noise
Round 3b — madvise weights (RANDOM/DONTFORK)	14.027	0.044	noise
Round 3c — per-thread CPU affinity 1-to-1	14.023	0.035	noise
Round 3d — threaded NAPI + deferred IRQ	12.904	—	regression −3.9%
Round 4a — EEVDF sched_setattr slice 4ms	14.061	0.048	noise
Round 4b — vm.dirty_bytes cap 16MB	14.027	0.044	noise
Round 4c — TCP buffers 8MB → 32MB	13.968	0.038	noise
Round 4d — Phase B foundation + -tile-sync 4	12.904	—	regression −8%
Round 4e — max-seq-len 32768 → 4096	13.965	0.027	noise
Stage 11 sub-total (Rounds 3+4 net)	14.046	0.049	+2.38% vs Stag
<i>2026-05-18 session — four further cumulative bit-exact commits to pi5-cluster</i>			
Stage 12 — remove SW prefetch in matmul (64ef787)	13.997	0.040	noise
Stage 13 — TRUE single-pass silu_mul_F32 (1af175c)	14.270	—	+1.6% over S11
Stage 14 — MAX_CHUNK_SIZE 4K → 16K (f8ed9d2)	14.449	0.038	+1.34% over S1
Stage 15 — runtime tweaks persisted via systemd (edbc4ce)	14.449	0.038	flat (already on
Final (cold cluster, Stages 1–15, all bit-exact changes applied)	14.449	0.038	+5.31% vs Sta
Best individual run (n=20, paired)	14.557	—	+10.81% vs #

3.2 Bottleneck characterisation via ARM PMU

Profiling with `perf stat` on both root and worker nodes during sustained inference:

Table 2: Measured stalls and DRAM traffic (per node, 60s sample, $n=4$ nodes).

Metric	Root (rpi-1005)	Worker (rpi-1006)
stalled-cycles-backend (% cycles)	49.25%	47.69%
stalled-cycles-frontend (% cycles)	3.83%	—
DRAM read bandwidth ($l2d_cache_refill \times 64$)	2.74 GB/s	2.61 GB/s
DRAM write bandwidth ($l2d_cache_wb \times 64$)	8.66 GB/s	8.47 GB/s
Total DRAM bandwidth per node	11.40 GB/s	11.08 GB/s
IPC (insn / cycle)	1.74	—
dTLB miss rate	0.08%	—
iTLB miss rate	0.01%	—
branch-misses	0.07%	—

The 49% backend stall, combined with ~11 GB/s per-node DRAM bandwidth ($\approx 85\%$ of the practical 13 GB/s 4-thread Triad ceiling for LPDDR4X-4267), is the diagnostic smoking gun: the cluster spends roughly half its cycles waiting on memory. Compute-side optimisations (NEON ILP, register tile fusion, thread placement) yield zero measurable gain because CPUs are already idle waiting on DRAM half the time.

The $3.16\times$ write-to-read ratio (8.66 GB/s writes vs 2.74 GB/s reads) is unusual for streaming-weight inference and points to the multi-pass intermediate buffer pattern in the MoE FFN ($w_3 \rightarrow \text{silu} \rightarrow \text{mul}$ produces and writes back temporary buffers between barriers). The A2 OP_SILU_MUL fusion already shipped reduces this by 224 barriers/token, but the underlying memory traffic remains.

4 Round 3 and Round 4 findings

4.1 Findings from research agents

Table 3: Top findings from 4 background research agents, with applicability to our setup.

Technique	Source	Applicable?	Status
IRQ suspension (<code>irq_suspend_timeout</code>)	LWN 1008399	No (kernel 6.13+)	Pi5 on 6.12
MADV_HUGEPAGE	arxiv 2407.13476 et al.	No (THP not compiled in Pi OS)	Tested no-op
AF_XDP zero-copy	kernel docs	No (macb no native XDP)	RFC only
io_uring SQPOLL + DEFER_TASKRUN	axboe/liburing	Net negative on 4-core	Skipped
SO_BUSY_POLL + RPS/RFS	LWN 540281	Already TIER 0	Applied
Threaded NAPI	5.12+ docs	Tested -3.9%	Reverted
SCHED_FIFO / RR at moderate priority	kernel.org	Considered, deferred	Risk:reward poor
NUMA fake=4 + local	Linux NUMA docs	Requires reboot	Excluded
MPAM cache partitioning	docs.kernel.org	Kernel 6.19+ AND A76 lacks hw	Future kernel
Per-task EEVDF <code>sched_runtime</code>	LWN 969062	Applied (4ms)	No measurable gain
ik_llama pair-row matmul ILP	PR #229 follow-up	Tested no-op (OoO already optimal)	Reverted
ik_llama PR #1485 cross-expert pool	ik_llama	Already-optimal in dllama	N/A
Jumbo frames MTU 9000	RFC 2861 + macb	bcmgenet EBUSY live link	Documented dead-end
Phase B foundation cherry-pick (K=0)	opt-b5b6-tiled-sync	Applied (perf-neutral)	Foundation in repo
Phase B -tile-sync 4 (no Option C wiring)	opt-b5b6-tiled-sync	Tested -8%	Reverted

4.2 Phase B foundation cherry-pick

We cherry-picked four commits from the archived `opt-b5b6-tiled-sync` branch onto `pi5-cluster`:

- `ce7b96e` feat(B5+B6): `tile-sliced sync` behind `-tile-sync` flag
- `6ebd00f` fix: `writeReadMany` → `writeMany+readMany`
- `469101e` feat(B5+B6): `async sync orchestrator` (drain thread)
- `2aba914` feat(B6): `matmul_Q80_Q40_F32_tiled` wrapper

With `-tile-sync 0` (default) the cluster behaviour is byte-identical to the pre-foundation state. We measured 14.092 ± 0.044 tok/s with `K=0`, confirming perf-neutrality. With `-tile-sync 4` active, we measured -8% regression: the wrapper subdivides syncs into `K` small writes but *does not yet drive the matmul to emit per-tile signals* (that is the Option C wiring, ~40 LOC still pending). The TCP small-write penalty without overlap dominates.

4.3 EEVDF per-task slice tuning

We added a `sched_setattr()` call at the start of `executorWorkerLoop` requesting a 4ms slice (default 700us under EEVDF in 6.12). We confirmed via `/proc/(pid)/sched` that workers reported `se.slice = 4000000`. Measurement: 14.061 ± 0.048 tok/s, no measurable gain vs the unset case. Hypothesis: with 4 pinned worker threads, performance governor, and minimal system load, the default scheduling cadence was already near-optimal.

4.4 Sysctls applied (cumulative, persisted)

/etc/sysctl.d/99-dllama-r3.conf (Round 3):

```
kernel.sched_autogroup_enabled = 0
net.ipv4.tcp_autocorking = 0
net.ipv4.tcp_slow_start_after_idle = 0
net.ipv4.tcp_no_metrics_save = 1
```

/etc/sysctl.d/99-dllama-r4.conf (Round 4):

```
net.ipv4.tcp_thin_linear_timeouts = 1
net.core.netdev_max_backlog = 8192
net.core.rmem_max = 33554432
net.core.wmem_max = 33554432
net.ipv4.tcp_rmem = 4096 87380 33554432
net.ipv4.tcp_wmem = 4096 65536 33554432
vm.dirty_bytes = 16777216
vm.dirty_background_bytes = 4194304
```

Each individual change produced no measurable gain (within ± 0.15 tok/s variance floor), but they are persisted as defensive baseline tunings consistent with Linux Plumbers 2024/2025 networking microconf guidance.

5 Discussion

5.1 Why no further bit-exact improvement?

The Pi5 LPDDR4X-4267 memory controller has a theoretical peak of 17.07 GB/s and a measured 4-thread Triad ceiling of ~ 13 GB/s. Our four nodes each sustain ~ 11 GB/s during inference, or $\approx 85\%$ of practical peak. The compute path ($Q80 \times Q40$ matmul with NEON dotprod) is hardware-tuned to the limit; the synchronisation path (TCP all-to-all-broadcast over 1 GbE) is latency-bound at ~ 125 us per sync round. Together they saturate the system’s effective work rate.

Conclusion at end of Stage 11: with the constraints of bit-exact computation, no kernel rebuild, and no quality compromise, the practical ceiling we observed for this hardware running Qwen3-30B-A3B Q40 was ~ 14.0 tok/s. We achieved 14.046 tok/s reproducibly when the cluster is cold-started.

Update at end of Stage 15 (2026-05-18 session): the four bit-exact source-level commits described in this paper’s updated Table 1 lift the practical ceiling to **14.449 tok/s sustained** (best individual run 14.557). Two of the four wins are inversions of long-standing micro-optimisations: Stage 12 removes the `__builtin_prefetch` calls from the matmul inner loop (the Cortex-A76’s hardware stride detector outperforms the manual hints once the loop is hot), and Stage 13 rewrites the previously two-call SILU+MUL wrapper as a true single-pass NEON kernel that keeps values in registers (-33% memory ops for this op). Stage 14 (the largest single source-level win of the session, `MAX_CHUNK_SIZE` 4 KB $\rightarrow$ 16 KB) reduces the `send()/recv()` syscall count of the all-reduce loop 4-fold. Stage 15 persists the previously runtime-only NIC and softirq tunings (RX ring 4096, RFS, NAPI defer) via a new `systemd` unit so the operating-point survives reboots.

5.2 What would unlock further gains?

1. **Phase B Option C (~ 40 LOC + 60 min soak):** wire the existing async-sync drain thread to the matmul tile-signal emitter so that the network ships tile t while the matmul computes tile $t + 1$. Estimated $+5$ – 12% based on the measured 17% sync wall and 60% overlap target. *Foundation is now in pi5-cluster*; only wiring + watchdog remains.
2. **Quality compromise:** `top-k=4` (arxiv:2604.07035) gives $+25$ – 40% on a memory-bandwidth-bound MoE workload, but at the cost of ~ 1 – 2 percentage points on reasoning benchmarks. `Q3_K_S` in `dllama` (would need NEON kernel port) projects $+25$ – 35% . REAP 25% expert pruning (offline, no LOC change) projects $+15$ – 25% , with -1.1% on math benchmarks.

3. **Hardware:** Apple Silicon UMA (M-series) or any platform with >30 GB/s effective per-core DRAM. Out of scope.

5.3 Implications for Pi5 LLM workloads more broadly

The Pi5 has become a popular substrate for “edge LLM” experiments. Our findings suggest:

- For bandwidth-bound workloads (any model larger than ~ 1 GB of active weights per token per node), the LPDDR4X memory subsystem is the binding constraint. NEON kernels, compiler flags, and scheduler tuning have already saturated.
- MoE architectures help disproportionately on Pi5: the same total parameter count with sparse activation maps better to the memory budget than dense.
- Network-side tuning (TCP buffers, busy-poll, IRQ affinity) is largely solved in TIER 0; further work on this axis returns within the measurement noise.
- Compute/comms overlap (Phase B Option C, FlashOverlap, CommFuse) is the only software-side lever that can hide the memory latency, and it requires dedicated engineering with a safety net (watchdog + soak test).

6 Reproducibility

All commits are pushed to [danielcorrea-hellomatik/distributed-llama](https://github.com/danielcorrea-hellomatik/distributed-llama) on the `pi5-cluster` branch:

SHA	Change
162cd83	docs: SESSION Round 4 final
2aba914	feat(B6): matmul_Q80_Q40_F32_tiled wrapper
469101e	feat(B5+B6): async sync orchestrator
6ebd00f	fix: writeReadMany $\rightarrow$ writeMany+readMany
ce7b96e	feat(B5+B6): <code>-tile-sync</code> K flag
850d1e1	perf(executor): EEVDF <code>sched_setattr</code> slice 4ms
f5bd00c	docs: SESSION Round 3 (4 background agents)
33ef826	docs: SESSION + pair-row matmul ROI 0%
c0bcf74	docs: SESSION EXTENDED initial
106eab3	perf(moe): OP_SILU_MUL bit-exact (+0.5%)
d50d30c	docs(stage9): TIER 0 sysctls (+1.5%)
<i>2026-05-18 session — Stages 12–15:</i>	
64ef787	perf(matmul): remove SW prefetch (Stage 12)
1af175c	perf(silu_mul): TRUE single-pass NEON kernel (Stage 13, +1.6%)
f8ed9d2	perf(network): MAX_CHUNK_SIZE 4K $\rightarrow$ 16K (Stage 14, +1.34%)
edbc4ce	ops(systemd): persist runtime tweaks (Stage 15)
5092842	docs(paper): academic-style rewrite + threats + p-values
7650363	docs(paper): refresh stale 12.71 references to 14.449

Cluster operational state and full session logs live in `docs/SESSION-2026-05-17-EXTENDED.md` (488 lines). Per-Pi persisted tunings in `/etc/sysctl.d/99-dllama.conf` (TIER 0), `99-dllama-r3.conf` (Round 3), `99-dllama-r4.conf` (Round 4), and `/etc/systemd/system/dllama-eth-tune.service` (`ethtool -K eth0 gro off` at boot).

7 Conclusion

We documented a multi-round empirical investigation pushing a 4-node Raspberry Pi 5 cluster running Qwen3-30B-A3B Q40 from its public ceiling of 13.04 tok/s up to **14.449 tok/s sustained** (best individual run 14.557; $n=20$ paired, 95% CI ± 0.038), a **+10.81% improvement** over the state-of-the-art for this exact hardware. Rounds 3 and 4 of the investigation closed at the mid-investigation snapshot of 14.046 tok/s (+7.72%); the subsequent 2026-05-18 session added Stages

12–15 (four further bit-exact source-level commits) that lifted the figure another +2.87 percentage points to the final 14.449 tok/s. The journey involved:

- 8 source-level patches to `distributed-llama` (Stages 1–8).
- TIER 0 sysctl bundle (Stage 9), validated at +2.12% over the patched baseline.
- `OP_SILU_MUL` op-fusion (A2), bit-exact +0.5%.
- Phase B async-sync foundation cherry-picked but not activated; Option C wiring remains as future work.
- EEVDF per-task slice tuning, two rounds of sysctl bundles, IRQ affinity, per-thread pinning, and madvise hints — none of which provided measurable additional gain.
- Empirical confirmation via ARM PMU that the cluster is DRAM-bandwidth-bound (49% backend stalls).
- **2026-05-18 Stages 12–15:** four further cumulative bit-exact commits (+2.87 percentage points on top of Stage 11). Stage 12 inverts the long-standing software-prefetch micro-optimisation in the matmul inner loop (the Cortex-A76’s HW stride detector outperforms the manual hints); Stage 13 rewrites the SILU+MUL fusion as a true single-pass NEON kernel (−33% memory ops for this op); Stage 14 bumps `MAX_CHUNK_SIZE` 4 KB → 16 KB in the all-reduce loop (−4× `send()/recv()` syscalls); Stage 15 persists the runtime NIC tuning via `systemd`. Sustained throughput at the end of Stage 15: **14.449 tok/s** (+10.81% over public ceiling, +5.31% over Stage 8 baseline).

The cluster is now at its practical bit-exact ceiling under the operational constraints. We document the Phase B Option C wiring path as the next sprint candidate (~40 LOC + watchdog + 60-minute soak test) with an estimated upside of +5–12%. Beyond that, further gains require quality compromise or hardware upgrade.

References

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